

PVC injectors 10 kg/h

with diaphragm non-return valve

545-0518, 545-1519, 545-05187

545-3518, 545-4519, 545-35187

Installation and operating instructions



Original installation and operating instructions

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**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Safety instructions

These installation and operating instructions contain important information for the user of the product:

- Technical data
- Instructions for commissioning and use
- Safety instructions

If you require further information, or if problems occur that are not described in detail in this manual, please contact Grundfos.

1.1 Symbols used in this document

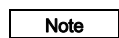
This manual contains the following standardised safety instructions about possible residual risks:

**Warning**

If these safety instructions are not observed, it may result in personal injury.

**Caution**

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

**Note**

Notes or instructions that make the job easier and ensure safe operation.

1.2 Safety advice for handling chlorine**1.2.1 Hazards of chlorine gas****Warning**

Toxic by inhalation.

Irritating to eyes, respiratory system and skin.

- Chlorine gas causes whooping cough, dyspnoea and lacrimation.
- Chlorine gas has a slight paralysing effect to the central nervous system.
- Concentrations of more than 10 ppm chlorine gas in the respiratory air mean an acute danger to life.
- Inhaling air with a high concentration of chlorine gas for a long time is lethal.

1.2.2 Hazards of liquid chlorine

- Liquid chlorine causes burns of skin.
- Liquid chlorine causes reddening and blistering of skin.

1.2.3 Rules of conduct

- Change chlorine containers only with gas mask.
- Enter contaminated rooms only with protective suit and compressed-air respirator.
- In case of flight wear gas mask, if possible. Observe wind direction.

1.3 Protective equipment**1.3.1 Personal protective equipment**

The operating authority of a chlorine gas dosing system has to provide respiratory equipment (full-sight gas mask), personally fitted, labelled by name, with an effective chlorine filter (B2P3) and at least 1 spare filter per gas mask, for each operator. The protective equipment must be stored easily accessible outside the chlorine rooms.

- For systems with chlorine drums at least 2 protective suits with compressed-air respirators must be available.

1.3.2 Obligations of the operating authority

- Introducing the operators to handling the protective equipment.
- The operators must use the personal protective equipment regularly, or carry out exercises with it at least every 6 months.
- The gas mask filters must be replaced regularly
 - after the expiry of the date of durability
 - at least 6 months after opening (note the opening date on the filter)
 - after contact with chlorine.
- Observe employing prohibition according to the local laws.

1.3.3 Storage of the protective equipment

- Outside the chlorine rooms.
- Well visible.
- Easily available at any time.
- Protected from dust and moisture.

2. General information

2.1 Correct usage

Injectors are used to inject Cl_2 , NH_3 and SO_2 gas into water in water treatment systems within the framework of the applications described in these operating instructions.

The media that may be used with each model are specified on the name plate.

2.2 Inappropriate usage

Operational safety is only guaranteed, if the product is used correctly. All operating methods conflicting with correct usage are not permitted, and lead to the expiry of all liability claims.



Warning

Unauthorised structural modifications to the product may result in serious damage to equipment and personal injury.

It is forbidden to open, modify, bridge, remove, bypass or disable components, especially safety equipment.

2.3 Qualification of the users

Based on his professional training and experience, the user must have a good knowledge of the operation of chlorine gas dosing systems. He must be so familiar with the relevant laws and regulations governing safety in the workplace and accident prevention, and with the recognised standards of good practice, that he is able to judge the safe working condition of a chlorine gas dosing system.

The user is responsible for operation and monitoring of the product at the installation site.

The responsibilities of the user:

- Read this manual before operating the product.
- Be trained by qualified Grundfos personnel in the operation of the product.
- Observe the relevant regulations governing safety in the workplace and accident prevention.
- Wear appropriate protective clothing in accordance with national regulations for the prevention of accidents when operating the system and handling chemicals.

2.4 Responsibilities of the operator

The operator of the product is responsible for the following:

- Ensure that this manual is kept clearly accessible in the immediate vicinity of the system for the entire service life.
- Meet the installation requirements specified by the manufacturer (environmental conditions, required water connections and fittings, electrical connection, protective pipe for dosing line if necessary, audible or optical warning device if necessary).
- Ensure that water lines and armatures are regularly checked, serviced and maintained.
- Obtain official approval for storing chemicals, if necessary.
- Train users in the operation of the system.
- Ensure that the regulations for the prevention of accidents are observed in the installation site.
- Provide all users and service personnel with protective clothing (e.g. face mask, gloves, protective apron) in accordance with the national regulations.

2.5 Maintenance and service personnel

The product may only be maintained and serviced by personnel authorised by Grundfos.

2.6 Warranty

The warranty period is 24 months from the date of delivery.

A guarantee claim in accordance with our general terms of sale and delivery is only valid if the following requirements are fulfilled:

- The product is used in accordance with the information within this manual.
- The product is not dismantled or handled incorrectly.
- The maintenance is carried out by authorised and qualified personnel.
- Only original spare parts are used for repairs during maintenance.
- Only parts permitted by Grundfos are used in the entire gas dosing system.

Typical wear parts are excluded from warranty, e.g.:

- gaskets, O-rings
- the injector nozzle, because the wear depends on the local water quality.

2.7 Transport and storage

- Transport carefully, do not throw!
- Store in a dry and cool place.

2.8 Unpacking

Observe when unpacking:

- The delivery must be complete.
- No humidity and no foreign matters should get into connections.
- Mount the injector as soon as possible after unpacking.

3. Technical data

3.1 General technical data

Dosing medium	Cl ₂ , NH ₃ , SO ₂ and CO ₂
Maximum suction capacity	10 kg/h Cl ₂ , 5 kg/h NH ₃ , 10 kg/h SO ₂ and 2.5 kg/h CO ₂
Motive water pressure	See section 3.3 Dimensioning of injectors
Counterpressure	
Permissible motive water temperature	+5 °C to +30 °C
Permissible ambient temperature	+5 °C to +40 °C
Gas dosing system	VGS 141

3.2 Connection data

Injector	Motive water connection inlet, outlet	Gas connection
545-0518	DN 25	optionally 10 x 14, 12 x 16, DN 10 or DN 15
545-3518		
545-05187		
545-35187		
545-1519	1 1/4" tube (1 1/4" NPT)	
545-4519		

3.3 Dimensioning of injectors

The criteria for the selection of an injector are suction power, counterpressure and motive water pressure (admission pressure).

Usually the required suction power and counterpressure is generally specified. Other parameters such as motive water pressure and water quantity per hour can be found in the dimensioning table.

Counterpressure [bar]	Motive water pressure [bar]	Water quantity [m ³ /h]	Injector
0.5	3	3.5	545-0518 or 545-1519 or 545-05187
1	4	4	
1.5	5	4.5	
2	6	4.8	
2.5	7	5.1	
3	8	5.5	
3.5	9	3.5	545-3518 or 545-4519 or 545-35187
4	10	3.7	
4.5	11	3.9	
5	12	4	
5.5	13	4.2	
6	14	4.3	
6.5	15	4.4	
7	16	4.5	

Note All data refers to drinking water quality motive water.

Example

To achieve a suction power of 10 kg/h Cl₂ with a counterpressure of 2.5 bar, a motive water pressure (admission pressure) of 7 bar and a water quantity of 5.1 m³/h is required.

3.4 Dimensions

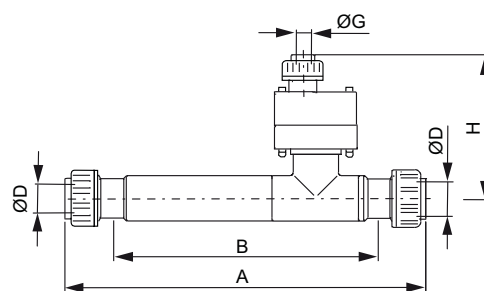


Fig. 1 Dimensioned drawing

Pos.	Description
A	Overall length
B	Fitting distance
H	Height

Injector	A [mm]	B [mm]	H [mm]
545-0518	400	300	155
545-3518			
545-05187			
545-35187	400	250	155
545-1519			
545-4519			

4. Structure and functioning

4.1 Functioning

By reducing the line cross-section with a nozzle, the motive water is accelerated and creates a negative pressure. The gas is drawn in by the negative pressure.

The gas is then whirled by the diffuser and mixed with the motive water in the downstream mixing section to produce operating water.

4.2 Structure

Components of the injector:

- Gas supply connection (1).
- Body with diaphragm non-return valve (2): The diaphragm non-return valve prevents water from entering the gas-conveying parts of the device.
- Motive water inlet connection (3).
- The nozzle (4) accelerates the motive water flow and creates a negative pressure.
- The diffuser with mixing section (5) mixes motive water with gas.
- Gas solution line connection = motive water outlet (6).

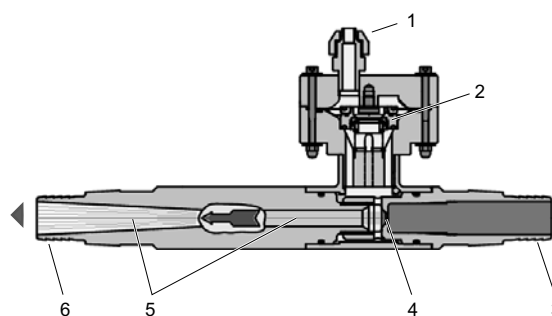


Fig. 2 Functioning scheme

5. Installation

The following conditions must be fulfilled to ensure proper functioning:

- Ensure sufficient motive water and motive water pressure (admission pressure). Fit a booster pump if necessary.
- Keep the solution line between the injector and the injection unit as short as possible.
- Use clean water, install a strainer if necessary.
- Choose the length and cross-section of the gas supply line according to the values proposed in the table below. For the calculation of the values, a pressure loss of 50 mbar was assumed, without considering a possible effect on the safety of the system.

Gas supply line [m]	Dosing quantity [g/h]							
	40	100	250	500	1000	2000	4000	10000
0	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 15
10	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 15
20	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
30	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
40	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
50	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 15
75	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 20
100	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 20

5.1 Mounting the injector

Caution Observe flow direction (arrow sticker).

Caution Mount and connect the injector free of stresses.

Fit the injector to the wall or other suitable location using the optional fixing material.

The optional fixing material (product number 95701439) includes two clamps, two M 6 x 60 screws and two S10 dowels.

5.2 Connecting the injector

1. Connect motive water supply line (pos. 3, fig. 2).
2. Connect gas solution line (6).
3. Connect gas supply line (1).

6. Startup

Caution

For commissioning, all components of the entire system must be ready for operation.
Observe the manuals of the other components.

6.1 Checks before start-up

Check the tightness of the entire system before start-up.



Warning
Only check for tightness when the entire system is ready for operation.
Observe the manuals of the other components.
Risk of gas leak.

6.1.1 Checking the gas solution line for tightness



Warning
The system must be switched off. All container valves must be closed.

1. Loosen the union nut (pos. 100, fig. 4) on the gas connection.
2. Disconnect the gas supply.
3. Open the motive water valve.
4. Open the injection unit isolating valve.
5. Switch on the motive water pump. If water escapes from between the motive water pump (pos. 3, fig. 3) and the injection unit (9), there is a leak in the line.
6. Switch off the motive water pump.
7. Close the motive water valve.
8. Close the injection unit isolating valve.
9. Repair the leak.
10. Repeat check for tightness.
11. If no water escapes, the gas supply can be reconnected.

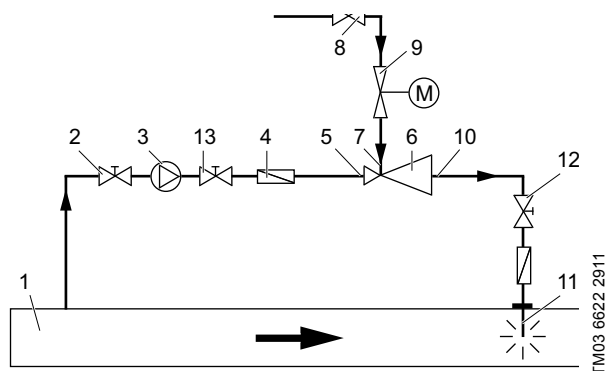


Fig. 3 System and components

Pos.	Description
1	Supply line
2	Isolating valve
3	Motive water pump
4	Non-return valve
5	Motive water inlet injector
6	Injector
7	Gas supply line
8	Safety pressure relief valve
9	Gas dosing system
10	Gas solution line (motive water outlet injector)
11	Injection unit

6.1.2 Checking the diaphragm non-return valve

1. Loosen the union nut (pos. 100, fig. 4) on the gas connection.
2. Disconnect the gas supply.
3. Open the motive water valve.
4. Open the injection unit isolating valve.
5. Switch on the motive water pump.
6. Briefly shut the injection unit isolating valve. If water escapes from the gas connection, the diaphragm non-return valve is defective.
7. Switch off the motive water pump.
8. Close the motive water valve.
9. Repair the diaphragm non-return valve.
10. Repeat the check. If no water escapes, the diaphragm non-return valve is working properly.
11. Reconnect the gas supply.

6.1.3 Checking the functioning of the injector

1. Loosen the union nut on the gas connection.
2. Disconnect the gas supply.
3. Open the motive water valve.
4. Open the injection unit isolating valve.
5. Switch on the motive water pump.
6. Place your thumb over the gas connection. If suction can clearly be felt, the injector is working correctly. If not, continue with the next point.
7. Check the admission pressure (water quantity).
8. Check the counterpressure.
9. Check the line length and cross-sections, see section 5. [Installation](#).
10. If the fault cannot be eliminated, contact the next Grundfos service centre.

7. Operation

Note

The injector is passive. Adjustment has to be made at other components of the system.

Read the manuals of the other components before start-up.

8. Maintenance



Warning

Before starting cleaning and maintenance work, switch off the entire system.
Danger of gas escape!



Warning

Before re-starting, check the tightness.
Danger of gas escape!

8.1 Intervals for cleaning and maintenance

Component	Normal use ¹⁾	Heavy duty ²⁾
Upstream dirt trap	every 12 weeks	every 8 weeks
Nozzles	every 12 weeks	every 8 weeks
Diaphragm non-return valve	every 6 months	every 6 months
Diaphragm non-return valve, piston, seat and diaphragm	every 6 months	every 4 months

¹⁾ Admission pressure up to 6 bar, high motive water quality (drinking water).

²⁾ Admission pressure more than 6 bar, poor motive water quality.

Note

In case of poor water quality, extreme pressure conditions, or faults, the maintenance intervals are shorter.

8.2 Maintenance kit

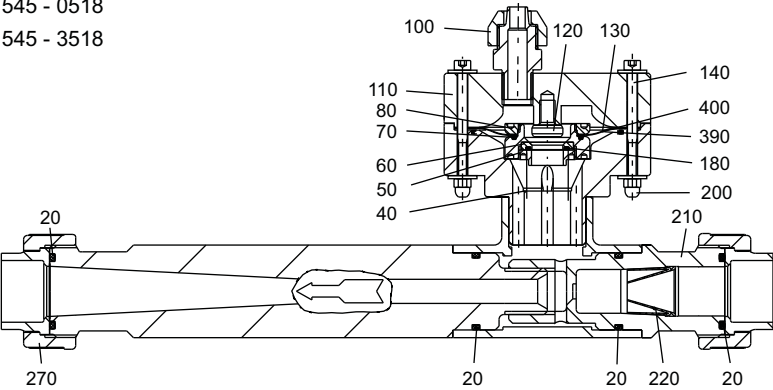
We recommend to keep maintenance kits in stock to ensure that worn parts can be replaced quickly when necessary.

The parts included in a maintenance kit are marked with an asterisk in the legend of the assembly drawing. See section [8.3 Assembly drawing](#).

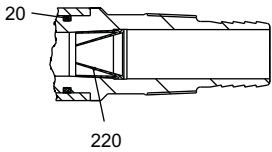
Maintenance kit	Product No.
for Cl ₂	96688738
for NH ₃ , SO ₂ and CO ₂	96729325

8.3 Assembly drawing

545 - 0518
545 - 3518



545 - 1519
545 - 4519



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Fig. 4 Assembly drawing

Pos.	Description
20	Four O-rings (33 x 40)*
40	Spring
50	Screw part
60	Seat*
70	O-ring (32 x 36)
80	Diaphragm ring
100	Union nut gas supply line
110	Top part
120	Piston*
130	Diaphragm*
140	Cylinder screw
160	O-ring (72 x 77)*
180	O-ring (17 x 22)*
200	Cap nut
210	Nozzle
220	Swirl part
390	Diaphragm*
400	Diaphragm*

* Included in maintenance kit

8.4 Cleaning the nozzle

Deposits in the nozzle prevent the injector from creating the required negative pressure, leading to a drop in suction capacity. We recommend to use hydrochloric acid for cleaning the nozzle.



Warning

Observe the regulations for the use of hydrochloric acid.

Caution Do not use sharp objects for cleaning.

1. Disconnect the gas supply.
2. Loosen the water line (on injectors with a tube connection, loosen the tube) and remove the injector.
3. Unscrew the nozzle (pos. 210, fig. 4). Leave the swirl part (220) in the nozzle.
4. Clean the nozzle with hydrochloric acid.
5. Reassemble and fit the injector in the reverse order.

8.5 Checking the diaphragm non-return valve

8.5.1 Checking the closing mechanism

1. Close the adjustment valve on the dosing system.
2. Loosen the union nut (100, fig. 4).
3. Disconnect the gas supply.
4. Briefly close the isolating valve behind the injector several times. If no water escapes, the diaphragm non-return valve is working properly. If water escapes, the piston and seat must be replaced.
5. Replace piston (120) and seat (60).
6. Open the isolating valve.
7. Connect the gas supply line.
8. Set the adjustment valve on the dosing system to the required dosing flow.

8.5.2 Replacing the piston

Do not use tools when screwing in the piston.

Caution If the piston surface is damaged, the closing mechanism may not work.

1. Shut down the system.
2. Close the isolating valve behind the injector.
3. Loosen the cap nut (pos. 200, fig. 4).
4. Remove the cylinder screws (140).
5. Remove the top part (90).
6. Unscrew the piston (120).
7. Screw in a new piston (by hand), replace O-ring (160) if necessary.
8. Slide the diaphragm disc (with diaphragm, seat, screw part and O-rings) onto the piston.
 - Turn the diaphragm disc so that the holes line up with the drill holes in the upper part.
 - Do not lose the spring (40).
9. Fit the top part and diaphragm disc.
10. Slightly tighten the screws in a criss-cross sequence.

8.5.3 Replacing the seat and diaphragm

1. Shut down the system.
2. Close the isolating valve behind the injector.
3. Loosen the cap nut (pos. 200, fig. 4).
4. Remove the cylinder screws (140).
5. Remove the top part (110).

Replacing the diaphragm

1. Unscrew the diaphragm ring (pos. 80, fig. 5).
2. Check O-ring (70), replace O-ring if necessary.
3. Remove diaphragm (130).
4. Insert a new diaphragm.

Caution Observe the mounting position.

5. Firmly tighten the diaphragm ring (80).

Replacing the seat

1. Unscrew the screw part (50).
2. Remove the seat (60) with O-ring.
3. Insert a new seat with O-ring.

Caution Observe the mounting position.

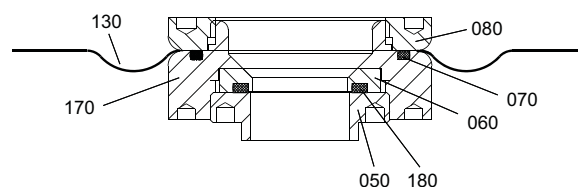


Fig. 5 Position of diaphragm

4. Tighten the diaphragm ring so that the seat can be pushed firmly yet smoothly over the piston.

8.5.4 Checking the spring

The spring is plastic-coated for protection against corrosive gases.

Caution Replace the spring, if the coating is damaged.

8.5.5 Assembling the diaphragm non-return valve

1. Slide the diaphragm disc (with diaphragm, seat, screw part and O-rings) onto the piston.

Caution Observe the mounting position.

2. Turn the diaphragm disc so that the holes line up with the drill holes in the upper part.
3. Insert the spring.
4. Fit the top part and diaphragm disc.
5. Fit the cylinder screws and slightly tighten them in a criss-cross sequence.
6. Check the closing mechanism.
7. If water escapes, re-tighten the diaphragm ring slightly.
8. If water still escapes, re-tighten the diaphragm ring again, until it closes tightly.

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9. Fault finding

Fault	Cause	Remedy
1. Maximum dosing flow is not achieved despite correctly selected injector.	a) Swirl part in the nozzle damaged or not present.	Insert a new swirl part.
	b) Counterpressure on the injector is too high (solution outlet too long or line cross-section too small).	Install the injector closer to the injection unit, observe the line cross-sections
	c) Motive water pressure or quantity not correctly set.	Check the settings using the table in section 3.3 Dimensioning of injectors .
	d) Dynamic pressure on the injector too low due to line loss.	Check length and cross-section of motive water and solution line, use the least possible number of elbows.
	e) Process water temperature higher than 30 °C.	Adjust process water temperature to approx. 20-25 °C.
	f) The dynamic pressure created in the injector is too low due to faulty gas supply.	Replace the gas supply line.

10. Disposal

This product and all its associated parts must be disposed of in an environmentally friendly manner. Use appropriate waste collection services. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

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